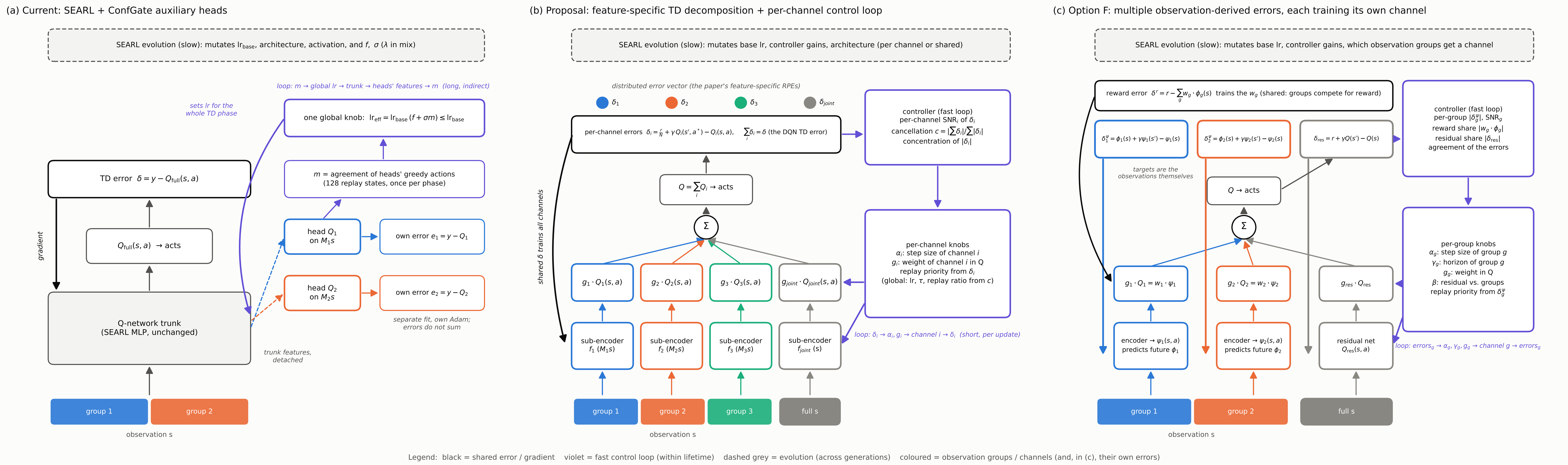
(a) Current: SEARL + ConfGate auxiliary heads
SEARL evolution (slow): mutates $\mathrm{lr}_{\mathrm{base}}$, architecture, activation, and f, σ (λ in mix)
loop: $m \to$ global lr $\to$ trunk $\to$ heads' features $\to m$ (long, indirect)
sets lr for the whole TD phase
one global knob: $\mathrm{lr}_{\mathrm{eff}} = \mathrm{lr}_{\mathrm{base}}(f + \sigma m) \le \mathrm{lr}_{\mathrm{base}}$
m = agreement of heads' greedy actions (128 replay states, once per phase)
TD error $\delta = y - Q_{\mathrm{full}}(s,a)$
gradient
$Q_{\mathrm{full}}(s,a) \to$ acts
head Q_1 on $M_1 s$
own error $e_1 = y - Q_1$
head Q_2 on $M_2 s$
own error $e_2 = y - Q_2$
separate fit, own Adam; errors do not sum
Q-network trunk (SEARL MLP, unchanged)
trunk features, detached
group 1
group 2
observation s
(b) Proposal: feature-specific TD decomposition + per-channel control loop
SEARL evolution (slow): mutates base lr, controller gains, architecture (per channel or shared)
distributed error vector (the paper's feature-specific RPEs)
δ_1
δ_2
δ_3
δ_{joint}
per-channel errors $\delta_i = \frac{r}{N} + \gamma Q_i(s', a^*) - Q_i(s,a)$, $\sum_i \delta_i = \delta$ (the DQN TD error)
controller (fast loop) per-channel SNR_i of δ_i cancellation $c = |\sum \delta_i| / \sum |\delta_i|$ concentration of $|\delta_i|$
shared δ trains all channels
$Q = \sum_i Q_i \to$ acts
Σ
$g_1 \cdot Q_1(s,a)$
$g_2 \cdot Q_2(s,a)$
$g_3 \cdot Q_3(s,a)$
$g_{joint} \cdot Q_{joint}(s,a)$
per-channel knobs α_i: step size of channel i g_i: weight of channel i in Q replay priority from δ_i (global: lr, τ, replay ratio from c)
sub-encoder f_1 ($M_1 s$)
sub-encoder f_2 ($M_2 s$)
sub-encoder f_3 ($M_3 s$)
sub-encoder f_{joint} (s)
loop: $\delta_i \to \alpha_i, g_i \to$ channel $i \to \delta_i$ (short, per update)
group 1
group 2
group 3
full s
observation s
(c) Option F: multiple observation-derived errors, each training its own channel
SEARL evolution (slow): mutates base lr, controller gains, which observation groups get a channel
reward error $\delta^r = r - \sum_g w_g \cdot \phi_g(s)$ trains the w_g (shared: groups compete for reward)
$\delta_1^\psi = \phi_1(s) + \gamma\psi_1(s') - \psi_1(s)$
$\delta_2^\psi = \phi_2(s) + \gamma\psi_2(s') - \psi_2(s)$
$\delta_{\mathrm{res}} = r + \gamma Q(s') - Q(s)$
controller (fast loop) per-group $|\delta_g^\psi|$, SNR_g reward share $|w_g \cdot \phi_g|$ residual share $|\delta_{\mathrm{res}}|$ agreement of the errors
targets are the observations themselves
$Q \to$ acts
Σ
$g_1 \cdot Q_1 = w_1 \cdot \psi_1$
$g_2 \cdot Q_2 = w_2 \cdot \psi_2$
$g_{res} \cdot Q_{\mathrm{res}}$
per-group knobs α_g: step size of group g γ_g: horizon of group g g_g: weight in Q β: residual vs. groups replay priority from δ_g^ψ
encoder $\to \psi_1(s,a)$ predicts future ϕ_1
encoder $\to \psi_2(s,a)$ predicts future ϕ_2
residual net $Q_{\mathrm{res}}(s,a)$
loop: $\mathrm{errors}_g \to \alpha_g, \gamma_g, g_g \to$ channel $g \to \mathrm{errors}_g$
group 1
group 2
full s
observation s
Legend: black = shared error / gradient violet = fast control loop (within lifetime) dashed grey = evolution (across generations) coloured = observation groups / channels (and, in (c), their own errors)